

A Feasibility Study on Tactile Enhancement of Mid-Air Ultrasonic Stimulation by Wrist Vibration

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Abstract. This study seeks a method to improve the perceived intensity of a mid-air ultrasonic tactile stimulus based on the perceptual phenomenon of tactile enhancement. We present a brief vibrotactile stimulus to the wrist before stimulating the user’s palm using a mid-air ultrasonic stimulus. The two stimuli have the same frequency. This arrangement is similar to the typical situation in which tactile enhancement occurs, except that the stimulation locations are close but different and the following stimulus is of mid-air ultrasonic type. Our method is validated by a user study demonstrating the perceived intensity of a mid-air ultrasonic stimulus is increased up to 1.7 times. These findings offer a new research direction to amend the most critical problem of current mid-air ultrasonic haptic devices, the weak perceptual strength, using a common wearable.

Keywords: Mid-air ultrasonic tactile stimulation · Tactile enhancement · Perceived intensity · Wearable

1 Introduction

The mid-air haptics technology allows users to perceive tactile sensations through the air without physical media. Since its inception using focused ultrasound [9, 4], mid-air haptics has been a significant milestone in the evolution of human-interactive technologies and applications; see Sec. 2.1 for a review. Although researchers introduced other attempts using laser [11, 14], the ultrasound-based technology is still the mainstream in mid-air haptics. However, it has two fundamental issues: the short energy transmission distance and the weak perceptual intensity, both of which hamper the technology’s usability considerably [23]. These two issues must be amended while preserving the unique advantage of mid-air haptics—free-hand operation.

In this paper, we explore methods to improve the perceptual strength of a mid-air focused ultrasonic stimulus while maintaining usability in existing mid-air applications. Our approach is based on the classical perceptual phenomenon of *tactile enhancement* [26]; see Sec. 2.2 for more details. As illustrated in Fig. 1 (left), if two vibrations are applied consecutively to the same position on the hand, the second vibration is perceived as stronger than presented alone. Applying this concept, we first present a short vibration to the wrist, e.g., using a wristband, and

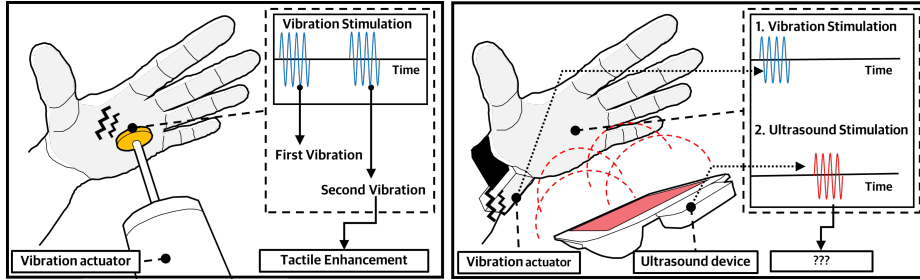


Fig. 1. Typical situation demonstrating tactile enhancement (left; reproduced from [26]) and our setup for mid-air ultrasound haptics (right).

then stimulate the palm using a mid-air ultrasound stimulus, as in Fig. 1 (right). This method differs from the typical conditions in which tactile enhancement is observed in two aspects: 1) the stimulation locations are different between the preceding and following stimuli, and 2) the stimulus types are different.

The rest of this paper describes the methods and results of a perceptual experiment conducted to investigate whether the proposed scenario indeed enhances the perceptual strength of an ultrasonic stimulus and quantify the extent of enhancement for different stimulus frequencies and amplitudes (Sec. 3 and 4). A discussion of the major results follows (Sec. 5), with conclusions (Sec. 6).

2 Related Works

2.1 Mid-air Haptics

Mid-air haptics using focused ultrasound enables free-hand interaction in the air and the creation of complex patterns on the palm [9], unlike other contact-required tactile feedback methods. This capability significantly expands the design space for haptic interaction and experiences [6], offering enhanced immersion in VR/AR content [15]. Focused ultrasound stimuli can be generated by controlling several variables, such as the sampling rate [8], focal point speed [19], and pattern size [1]. Additional parameters are also included as variables depending on the modulation method used. For instance, the modulation frequency is a critical parameter in the amplitude modulation technique [25].

Researchers developed many applications of mid-air haptics. For example, Pan et al. designed mid-air haptic stimuli to transmit Braille, underscoring its potential for tactile communication [18]. Kamigaki et al. delivered heat and vibrational sensations to hands wearing gloves using ultrasound [12]. Brist et al. conducted a study delving into how haptic interactions influence the interpretation of emotional stimuli, thereby suggesting the utility of mid-air haptics in conveying and mediating emotions [17]. In art exhibitions, mid-air haptics allowed visitors to interact with artwork through haptic patterns, enhancing the multisensory experiences [27]. Research was also conducted on haptic design,

such as user-defined haptic sensations [2] and design tools for effective mid-air haptic experiences [23].

Researchers are well aware that the weak intensity of mid-air ultrasound tactile experiences negatively affects usability and information communication [20, 23]. To overcome the limitation, Driller et al. explored varying the frequency or duration of a mid-air stimulus [5], and Frier et al. altered the wave modulation method of ultrasound [7]. Compared with these methods, our idea leaves the device and mid-air stimuli intact while appending a common wearable, a wristband, to provide accompanying tactile stimuli.

2.2 Tactile Enhancement

Tactile enhancement refers to a phenomenon where a sound following another sound is perceived louder without increasing its physical intensity. This effect is observed across various types of auditory stimuli and is affected by many factors, such as timing, frequency, and intensity [28, 29]. Tactile enhancement could increase the perceived loudness of auditory stimulus up to 10 dB [28].

Verrillo et al. discovered a similar effect for touch [26]. They provided participants with consecutive vibrotactile stimuli, similar to the auditory stimuli used in the loudness enhancement studies, and found that the enhancement phenomenon also occurs in the tactile modality. It was observed that tactile enhancement occurred most significantly when two vibrations of the same frequency were experienced consecutively with a short inter-stimulus interval (ISI). Furthermore, when the intensity of the first stimulus was 10 dB stronger than that of the second stimulus, the perceived intensity of the second stimulus increased approximately twofold. Here, the two tactile stimuli were of the same type and frequency and stimulated the same body site.

3 Methods

We conducted a perceptual experiment to explore the potential of tactile enhancement of ultrasonic mid-air tactile stimuli by contact tactile stimuli. This experiment received ethical approval from the Institutional Review Board at POSTECH (No. PIRB-2023-E037).

3.1 Participants

Twenty volunteers (9 females and 11 males; 20–32 years old, average 24.5) participated in the experiment. All participants had normal or corrected-to-normal sensory ability. Each participant provided informed consent before starting the experiment and received KRW 15,000 after the experiment. The entire experiment took approximately one hour.



Fig. 2. Mid-air ultrasonic haptic interface in a box (left). Participant conducting the experiment (middle). Vibration actuator attached to the wrist (right). Both mid-air and contact vibrations are applied to the same hand: palm and palmar wrist, respectively.

3.2 Stimuli

We used two types of tactile stimuli. One was a contact type generated by a regular voice-coil actuator (HapCoil-One, Actronika) stimulating the wrist (Fig. 2, right). The other was a mid-air type produced by a commercial ultrasound haptic device (STRATOS™ Explore Development Kit, Ultraleap) (Fig. 2, left). Participants perceived the two types of stimuli on the same hand (Fig. 2, middle) to evaluate the extent of tactile enhancement.

Tactile enhancement between two consecutive vibrations is maximized if they share the same frequency [26]. We selected two frequencies (50 and 200 Hz) for this experiment, which primarily stimulate the RA1 and RA2 mechanoreceptive channels, respectively [3]. Tactile enhancement also depends on the intensities of the two vibrations [26]. Thus, we combined the two frequencies factorially with three intensity levels (NONE, WEAK, and STRONG) for contact-type vibrations. For NONE, no contact vibration was presented, so participants perceived only mid-air tactile stimuli. The specific vibration amplitudes were determined by a pilot experiment, as the perceived intensity of a vibration also depends on its frequency [10, 13]. Using the method of adjustment, six haptics experts found the vibration amplitudes that maintained the same perceived intensities at the two frequencies for the two levels of WEAK and STRONG. The average values were 2.9 and 8.8 G at WEAK and 1.3 and 6.2 G at STRONG, respectively. These contact vibration amplitudes were used in the main experiment.

The mid-air vibration signal was commanded as follows (Fig. 3):

$$p(t) = \frac{I_m}{2} (1 + \sin(2\pi f_m t)) \sin(2\pi f_u t), \quad (1)$$

where I_m is the normalized amplitude, f_m is the modulation frequency, and f_u is the carrier frequency of ultrasound (40 kHz). f_m corresponds to the frequency of focal point vibrations employing amplitude modulation, which is what the user perceives [8]. The mid-air stimuli were presented in two intensity levels, WEAK

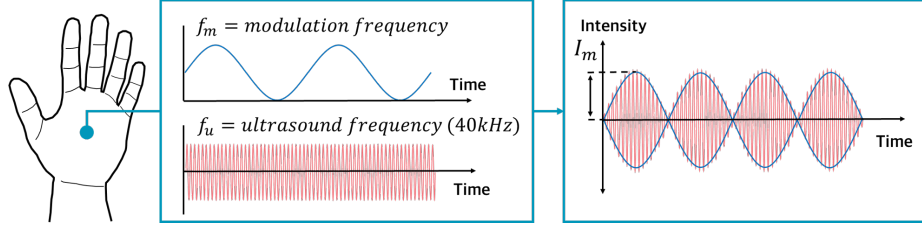


Fig. 3. Ultrasound stimulation method using amplitude modulation.

and **STRONG**. They were implemented by commanding the ultrasonic haptic device using 0.2 and 0.5 of its maximum intensity. These values are ratios and denoted using a unit AU (arbitrary unit) hereafter. Their perceptual effects are characterized by a follow-up experiment (Sec. 5.2). The distance between the stimulation point and the device was maintained at 20 cm.

The last critical parameters were the stimulus durations and the inter-stimulus interval (ISI). Previous research on tactile enhancement showed that a shorter ISI resulted in a more pronounced effect [26]. In this study, we set the duration of both contact and mid-air vibrations to 300 ms and the ISI to 100 ms, as illustrated in Fig. 4, right.

Table 1 summarizes the 12 experimental conditions of this study. Note that no preceding contact stimulus was presented in Conditions 1–4, the baselines.

Table 1. Experimental conditions.

No.	1	2	3	4	5	6	7	8	9	10	11	12
Frequency (Hz)	50	50	200	200	50	50	50	50	200	200	200	200
Contact Intensity	NO	NO	NO	NO	WK	SR	WK	SR	WK	SR	WK	SR
Mid-air Intensity	SR	WK	SR	WK	SR	SR	WK	WK	SR	SR	WK	WK

3.3 Task and Procedure

Participants wore the mechanical actuator that provided contact tactile stimuli on their wrists. A rectangular box enclosing the ultrasonic haptic device was placed beside the participants. The mid-air device was positioned 20 cm below the aperture in the box’s top cover. Participants were instructed to place their left palms over the aperture to perceive mid-air tactile stimuli (Fig. 2). Participants controlled the GUI (Fig. 4) using their right hands while focusing visually on the screen. They also wore headphones that played white noise to block out sounds generated by the actuators.

The experiment consisted of eight sessions. The first session was considered training to help participants become familiar with the experimental procedure.

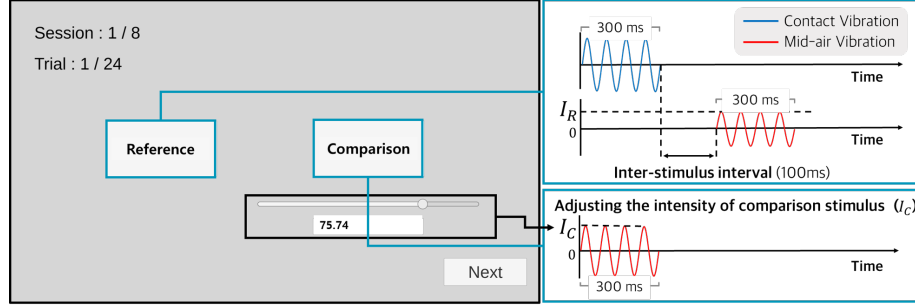


Fig. 4. Experimental program interface (left). When a button (reference or comparison) was pressed, the corresponding stimulus (mid-air + contact or mid-air only) was generated (right). The slider below the comparison button allowed participants to adjust the comparison stimulus intensity.

This first session was excluded from the data analysis. Each session included 24 trials, presenting the 12 experimental conditions twice in a random order. Hence, each experimental condition was evaluated using the data of 14 repetitions (7 sessions $\times$ 2 times).

In each trial, participants experienced the consecutive stimulus pair, a contact tactile stimulus followed by a mid-air stimulus after the ISI, assigned to the trial by clicking the “reference” button on the GUI. They also perceived a mid-air tactile stimulus with a varying amplitude by clicking the “comparison” button. The amplitude was adjusted in AU between 0 and 1 using a slider on the GUI, which allowed participants to adjust their responses easily and precisely. Participants repeated to perceive the reference and comparison stimuli while changing the comparison stimulus amplitude until the two stimuli were perceived as equally strong. This procedure followed the method of adjustment, and the initial comparison amplitude was randomly set in every trial. After completing the match, participants clicked the “Next” button to proceed to the next trial.

3.4 Data Analysis

We evaluated the extent of tactile enhancement by comparing the intensity I_R of the reference mid-air stimulus in the stimulus pair with the intensity I_C of the comparison mid-air stimulus at which participants perceived as strong as the reference (contact + mid-air) stimulus. I_R and I_C are graphically illustrated in Fig 4, right. Each experimental condition was assessed 14 times, and we averaged the 14 values of I_C to obtain the threshold estimate $\overline{I_C}$ of the condition.

4 Results

Fig. 5 presents all thresholds of $\overline{I_C}$ for equivalent perceived intensity estimated under the 12 experimental conditions. One-way repeated-measures ANOVA showed

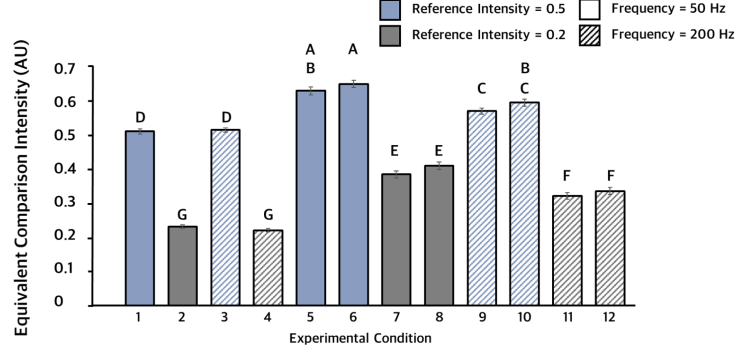


Fig. 5. Experiment results. Error bars represent standard errors. Experimental conditions grouped with the same letter did not have significant differences by SNK tests. Conditions not grouped with any letter had significant differences.

that the experimental condition was a significant factor for $\overline{I_C}$ ($F(11, 209) = 25.17, p < 0.0001$). The SNK tests revealed that all combined stimuli resulted in significantly higher $\overline{I_C}$ than the corresponding reference mid-air stimuli; compare Experimental Condition 1 with 5 and 6, 2 with 7 and 8, 3 with 9 and 10, and 4 with 11 and 12 (Table 1). We also conducted a three-way repeated-measures ANOVA with independent factors of vibration frequency, contact stimulus intensity, and mid-air stimulus intensity. All three factors were significant for $\overline{I_C}$ ($F(1, 19) = 10.97, p = 0.0037$; $F(2, 38) = 65.53, p < 0.001$; $F(1, 19) = 319.72, p < 0.001$).

To better understand the enhancement effects, we computed the difference from the reference intensity to the intensity threshold, $\Delta I = I_C - I_R$, where I_R is a constant (0.5 or 0.2 AU), for every response. Three-way repeated-measures ANOVA showed that ΔI was significantly affected by all three factors of vibration frequency ($F(1, 19) = 10.97, p = 0.0037$), contact stimulus intensity ($F(2, 38) = 65.53, p < 0.001$), and mid-air stimulus intensity ($F(1, 19) = 13.27, p = 0.0017$). We conducted the SNK tests for the three significant factors, and their results are depicted in Fig. 6. Using the vibration frequency of 50 Hz instead of 200 Hz resulted in significantly greater increases in perceived intensity. The mid-air stimuli with lower intensity obtained significantly more pronounced enhancements than those with higher intensity. Finally, the enhancement effect was significant when the contact tactile stimulus was present, but no significant differences were found between the two contact stimulus intensities (WEAK and STRONG).

5 Discussion

5.1 Effects of Preceding Tactile Stimulus

In the absence of a preceding contact tactile stimulus, the perceived intensity differences between the mid-air stimuli experienced by participants averaged

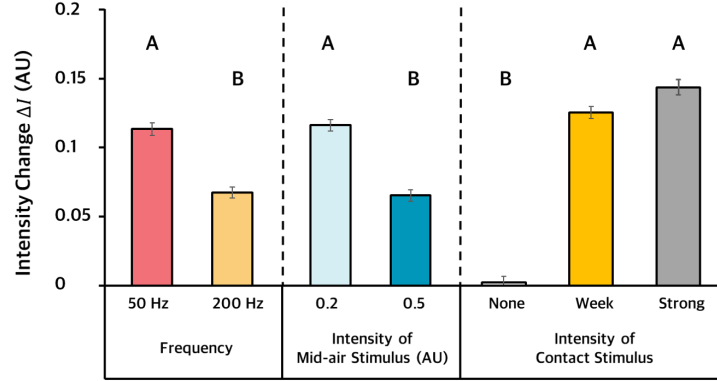


Fig. 6. Main factor effects for the perceived intensity difference ΔI . The factor means indicated by different letters had significant differences by the SNK tests.

0.0018 AU, which did not significantly differ from zero. In contrast, when a leading contact tactile stimulus was present, participants reported the average intensity 0.14 AU stronger than the actual intensities of mid-air stimuli. This result provides strong evidence that presenting a contact tactile stimulus on the wrist shortly followed by a mid-air tactile stimulus on the palm enhances the perceptual strength of the mid-air stimulus.

Increasing the contact stimulus intensity improved the enhancement effect, but its extent did not reach a significant level in the post-hoc tests (Fig. 6). We may need to test a wider range of contact stimulus intensity for more definite clarification of its effect.

As depicted in Fig. 6, the overall increase in perceived intensity was higher when the vibration frequency was 50 Hz than 200 Hz with statistical significance. Thus, tactile enhancement was more apparent for mid-air vibrations mediated through the RA-1 channel than the RA-2 channel.

We also analyzed participants' comments and interviews. Participants who perceived a higher intensity increase in the 50-Hz conditions than in the 200-Hz conditions commented that they felt the lower-frequency contact vibrations better spread across the palm (17 Ps). Participants with smaller hands reported that after experiencing contact tactile vibrations, their entire palms became sensitive, leading to a stronger perception of mid-air haptic stimuli (6 Ps). Participants who showed less enhancement effect mentioned that the vibration did not fully transmit to their palms (8 Ps). The degree of enhancement seems to relate to how well a contact vibration generated at the wrist propagates to the palm; low-frequency vibrations are generally better transmitted through the palm [24]. Control of how contact vibrations are transmitted from the wrist to the palm might be a key for the effective enhancement of mid-air ultrasound stimuli.

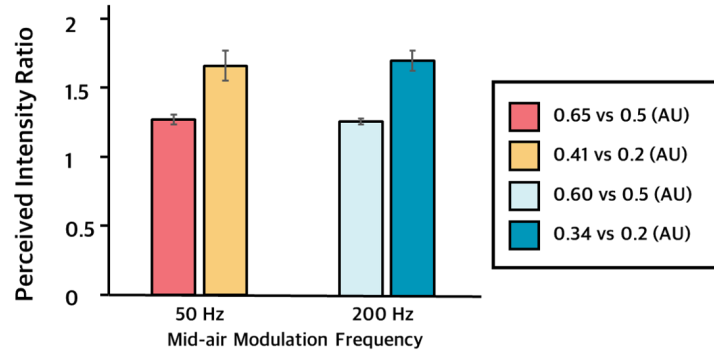


Fig. 7. Perceived intensity ratios validating the extents of tactile enhancement.

5.2 Characterization of Perceived Intensity Increases

The present experiment demonstrated that the intensity of a preceding contact tactile stimulus has the most significant impact on increasing the perceived intensity of a following mid-air tactile stimulus. This tactile enhancement effect was the most apparent in experimental conditions 6, 8, 10, and 12 (Table 1). In these conditions, the mid-air tactile stimuli with the program intensities of 0.5, 0.2, 0.5, and 0.2 AU were enhanced to those with 0.65, 0.41, 0.60, and 0.34 AU, respectively. However, the unit used here is an arbitrary proportional constant for programming, and it does not bear direct physical or perceptual meanings. Physical measurement of the mid-air vibration signals stimulating the user's palm is a straightforward solution to understanding their physical characteristics, but it is technically very challenging.

Instead, we took a perceptual approach by conducting another experiment for magnitude ratio scaling [16]. Five haptics experts were presented with four pairs of mid-air vibrations: (0.50, 0.65 AU) and (0.20, 0.41 AU) for 50 Hz, and (0.50, 0.60 AU) and (0.20, 0.34 AU) for 200 Hz. They responded with a ratio of how many times the second stimulus felt stronger than the first stimulus for each pair. These ratios represent the greatest gains in perceived intensity obtained in our experiment owing to tactile enhancement.

The results are presented in Fig. 7. For the four experimental conditions 6, 8, 10, and 12, the ratios in perceived intensity were 1.23, 1.67, 1.27, and 1.71. The average perceived intensity gain was 1.69 for the WEAK mid-air vibrations (0.2 AU) and 1.25 for the STRONG ones (0.5 AU).

5.3 Limitations and Future Work

Significant effects were obtained by our idea of enhancing the perceived intensity of mid-air ultrasonic haptic stimulation using tactile enhancement by a preceding wrist stimulation. This initial finding presents a few immediate issues that need to be addressed in future work.

First, we used a vibration actuator on the wrist to deliver a strong vibration to the palm for tactile enhancement. A better approach would be to deliver these preceding vibrations without the need for extra devices. For instance, they might be replaced with shear shock waves generated by ultrasonic tactile stimulation [22, 21]. Such methods can simplify the setup and improve user comfort.

Second, the two haptic devices used in our study have different actuation delays. We need to calibrate the ISI by measuring the exact moments at which the two stimuli begin to stimulate the skin. Then, we can quantify the effects of the ISI and the relative intensity difference between the preceding and following stimuli. Such calibrated results would be applicable to other devices.

Third, we plan to extend our method to spatiotemporal pattern rendering using a ultrasonic mid-air haptic device. Tactile enhancement using a preceding wrist vibration may improve the recognition performance of such patterns.

6 Conclusions

Mid-air ultrasonic haptic interfaces have immense potential, but their weak perceptual output has blocked their way to wider dissemination. In this paper, we have presented a method that increases the perceptual strength of a mid-air ultrasound stimulus while preserving the hand's free use. In this method, a vibrotactile stimulus stimulates the user's wrist, and then an ultrasound stimulus is focused on the palm after an inter-stimulus interval in order for tactile enhancement. It was demonstrated that our method can increase the perceived intensity of a mid-air stimulus up to 1.7 times. The enhancement effect was pronounced if the vibration frequency was low to stimulate the RA-1 channel and the mid-air stimulus had a low intensity. The dependence on the contact stimulus intensity requires more evidence for verification.

This work merely proves the feasibility of our approach, and many more research items await. Our approach is still adaptable for enhancing the perceptual intensity of ultrasonic tactile stimuli. We hope it could eventually contribute to improving the usability of mid-air ultrasonic haptic interfaces.

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